

```
In [2]: import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

plt.style.use("ggplot")
```

```
In [3]: df = pd.read_csv(r"C:\Users\priya\OneDrive\Desktop\task 5\Titanic-
Dataset.csv")
df.head()
```

Out[3]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	

```
In [4]: df.shape
```

Out[4]: (891, 12)

```
In [5]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
#   Column      Non-Null Count  Dtype
---  -
0   PassengerId  891 non-null    int64
1   Survived     891 non-null    int64
2   Pclass       891 non-null    int64
3   Name         891 non-null    object
4   Sex          891 non-null    object
5   Age          714 non-null    float64
6   SibSp        891 non-null    int64
7   Parch        891 non-null    int64
8   Ticket       891 non-null    object
9   Fare         891 non-null    float64
10  Cabin        204 non-null    object
11  Embarked     889 non-null    object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
```

```
In [6]: df.describe()
```

Out[6]:

	PassengerId	Survived	Pclass	Age	SibSp
count	891.000000	891.000000	891.000000	714.000000	891.000000
mean	446.000000	0.383838	2.308642	29.699118	0.523008
std	257.353842	0.486592	0.836071	14.526497	1.102743
min	1.000000	0.000000	1.000000	0.420000	0.000000
25%	223.500000	0.000000	2.000000	20.125000	0.000000
50%	446.000000	0.000000	3.000000	28.000000	0.000000
75%	668.500000	1.000000	3.000000	38.000000	1.000000
max	891.000000	1.000000	3.000000	80.000000	8.000000

```
In [7]: df.isnull().sum()
```

```
Out[7]: PassengerId      0
        Survived         0
        Pclass          0
        Name            0
        Sex             0
        Age            177
        SibSp           0
        Parch           0
        Ticket          0
        Fare            0
        Cabin          687
        Embarked        2
        dtype: int64
```

```
In [8]: df.duplicated().sum()
```

```
Out[8]: np.int64(0)
```

```
In [10]: df["Age"] = df["Age"].fillna(df["Age"].median())
```

```
In [11]: df["Embarked"] = df["Embarked"].fillna(df["Embarked"].mode()[0])
```

```
In [12]: df = df.drop("Cabin", axis=1)
```

```
In [13]: df.isnull().sum()
```

```
Out[13]: PassengerId      0
        Survived         0
        Pclass          0
        Name            0
        Sex             0
        Age             0
        SibSp           0
        Parch           0
        Ticket          0
        Fare            0
        Embarked        0
        dtype: int64
```

```
In [14]: df["Survived"].value_counts()
```

```
Out[14]: Survived
0      549
1      342
Name: count, dtype: int64
```

```
In [15]: df["Sex"].value_counts()
```

```
Out[15]: Sex
male      577
female    314
Name: count, dtype: int64
```

```
In [16]: df["Pclass"].value_counts()
```

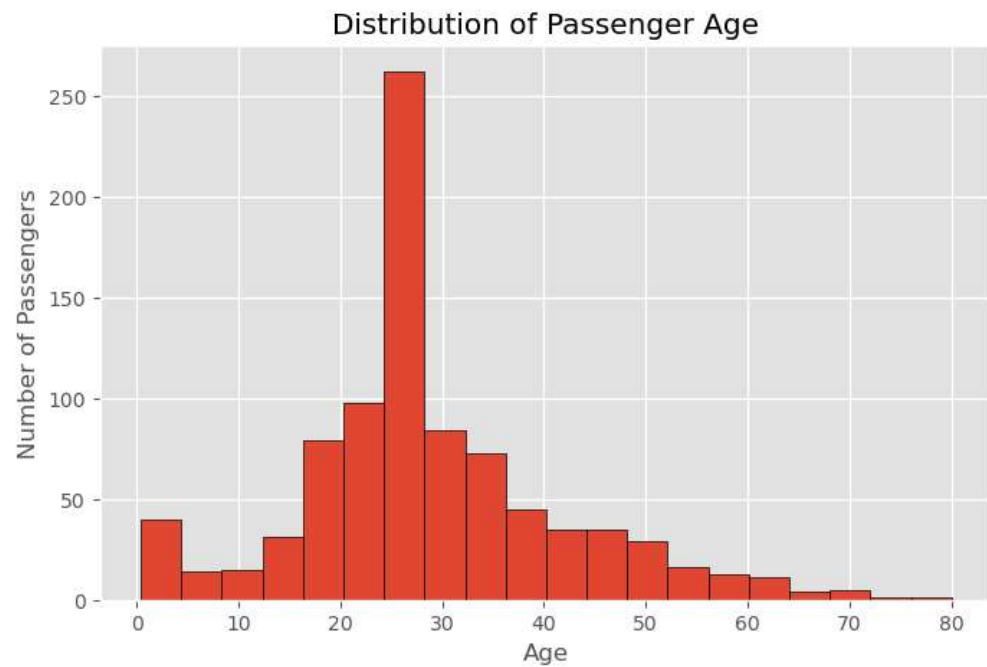
```
Out[16]: Pclass
3      491
1      216
2      184
Name: count, dtype: int64
```

```
In [17]: df["Embarked"].value_counts()
```

```
Out[17]: Embarked
S      646
C      168
Q       77
Name: count, dtype: int64
```

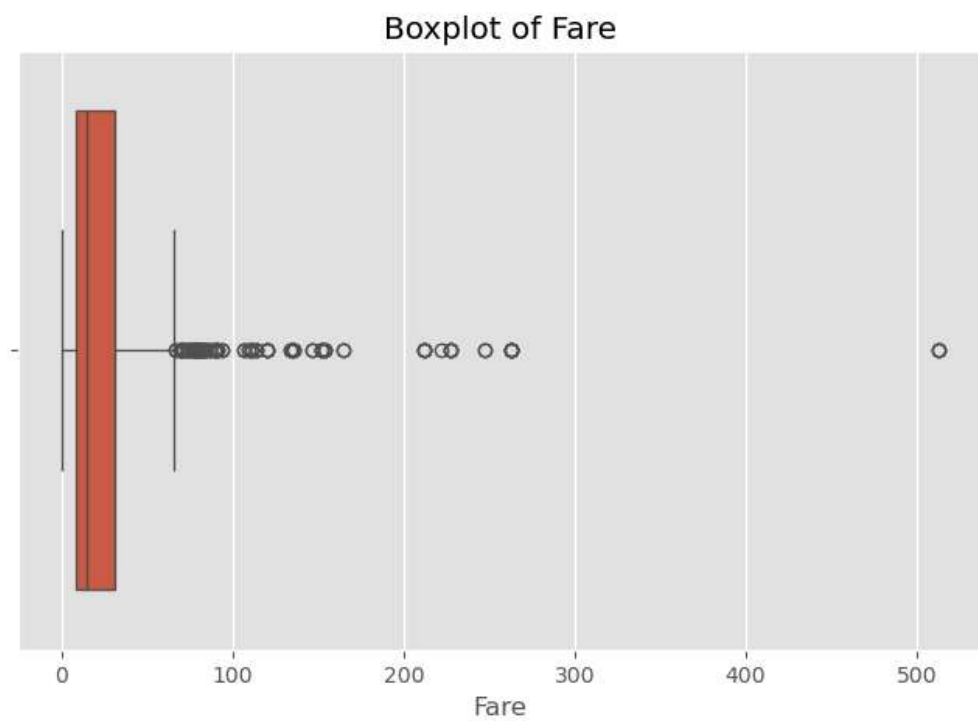
```
In [18]: plt.figure(figsize=(8,5))
plt.hist(df["Age"], bins=20, edgecolor="black")
plt.title("Distribution of Passenger Age")
plt.xlabel("Age")
plt.ylabel("Number of Passengers")

plt.show()
```



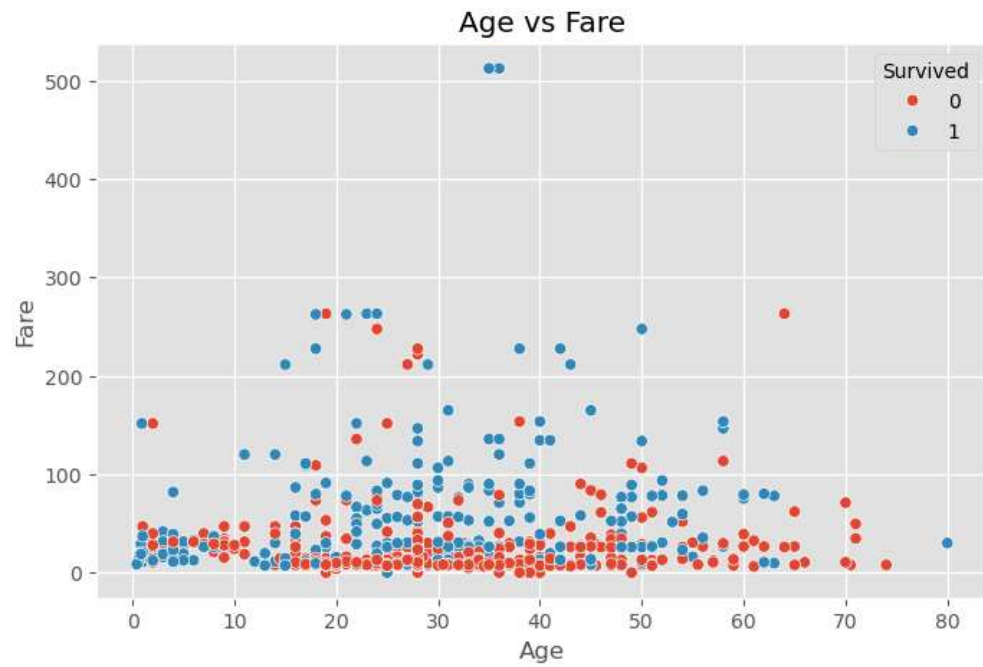
```
In [19]: plt.figure(figsize=(8,5))
sns.boxplot(x=df["Fare"])
plt.title("Boxplot of Fare")

plt.show()
```



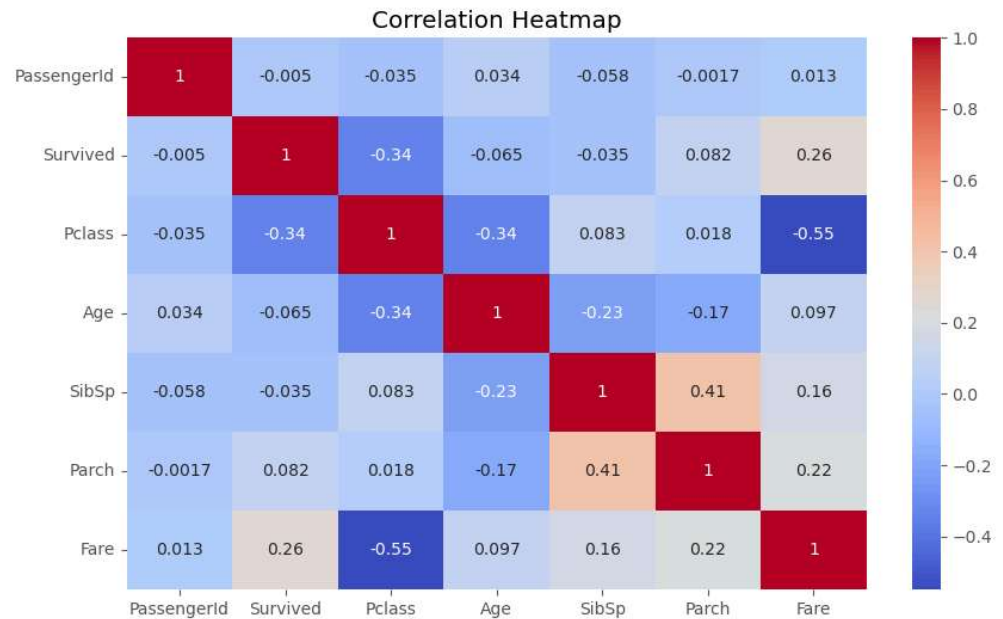
```
In [20]: plt.figure(figsize=(8,5))
sns.scatterplot(data=df, x="Age", y="Fare", hue="Survived")
plt.title("Age vs Fare")
plt.xlabel("Age")
plt.ylabel("Fare")

plt.show()
```



```
In [21]: plt.figure(figsize=(10,6))
sns.heatmap(df.select_dtypes(include="number").corr(),
            annot=True,
            cmap="coolwarm")
plt.title("Correlation Heatmap")

plt.show()
```




```
In [22]: sns.pairplot(df[["Age", "Fare", "Pclass", "SibSp", "Parch", "Survived"]],
          hue="Survived")

plt.show()
```



```
In [ ]: ### Findings
```

- The dataset contains 891 passengers and 12 variables.
- Missing values are present in Age, Cabin, and Embarked.
- Most passengers were aged 20–40 years.
- Most passengers traveled in third class.
- Females had a much higher survival rate than males.
- First-class passengers survived more frequently than those in lower classes.
- Higher ticket fares were generally associated with higher survival rates.
- The Fare variable contains several outliers.
- There were more deaths than survivors overall.